PRELIMINARY — NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TESTING, MOTION, OR ENERGIZATION

PROJECT BUTTON HR-V0 ELECTRICAL V3 OBSERVATION-INTEGRATED CANDIDATE

P1.15 limiter chain plus exact R202/R204 field and compute observation interfaces; zero safety credit.

01 External listed sources and DC boundaries

File: 01_external_sources.kicad_sch

02 Dual-channel E-stop and RESET eligibility

File: 02_estop_eligibility.kicad_sch

03 Distinct ARM and watchdog-gated SR1 supply

File: 03_arm_watchdog_eligibility.kicad_sch

04 Contactor coils, mirror contacts and EDM

File: 04_contactor_edm.kicad_sch

05 Redundant actuator-power interruption

File: 05_actuator_interruption.kicad_sch

06 Protected actuator branches and current-limiter carriers

File: 06_branches_and_limiters.kicad_sch

07 Independent watchdog power, controller and drivers

File: 07_watchdog_control.kicad_sch

08 Calculated dual-channel 24 V watchdog feedback

File: 08_watchdog_feedback_interface.kicad_sch

09 Compute and control terminals

File: 09_compute_and_control_terminals.kicad_sch

10 U2D2, DXL star, actuator ports and bonding boundary

File: 10_actuator_interfaces.kicad_sch

11 Watchdog PCB external connectors

File: 11_watchdog_pcb_connectors.kicad_sch

12 Watchdog PCB test access

File: 12_watchdog_pcb_test_access.kicad_sch

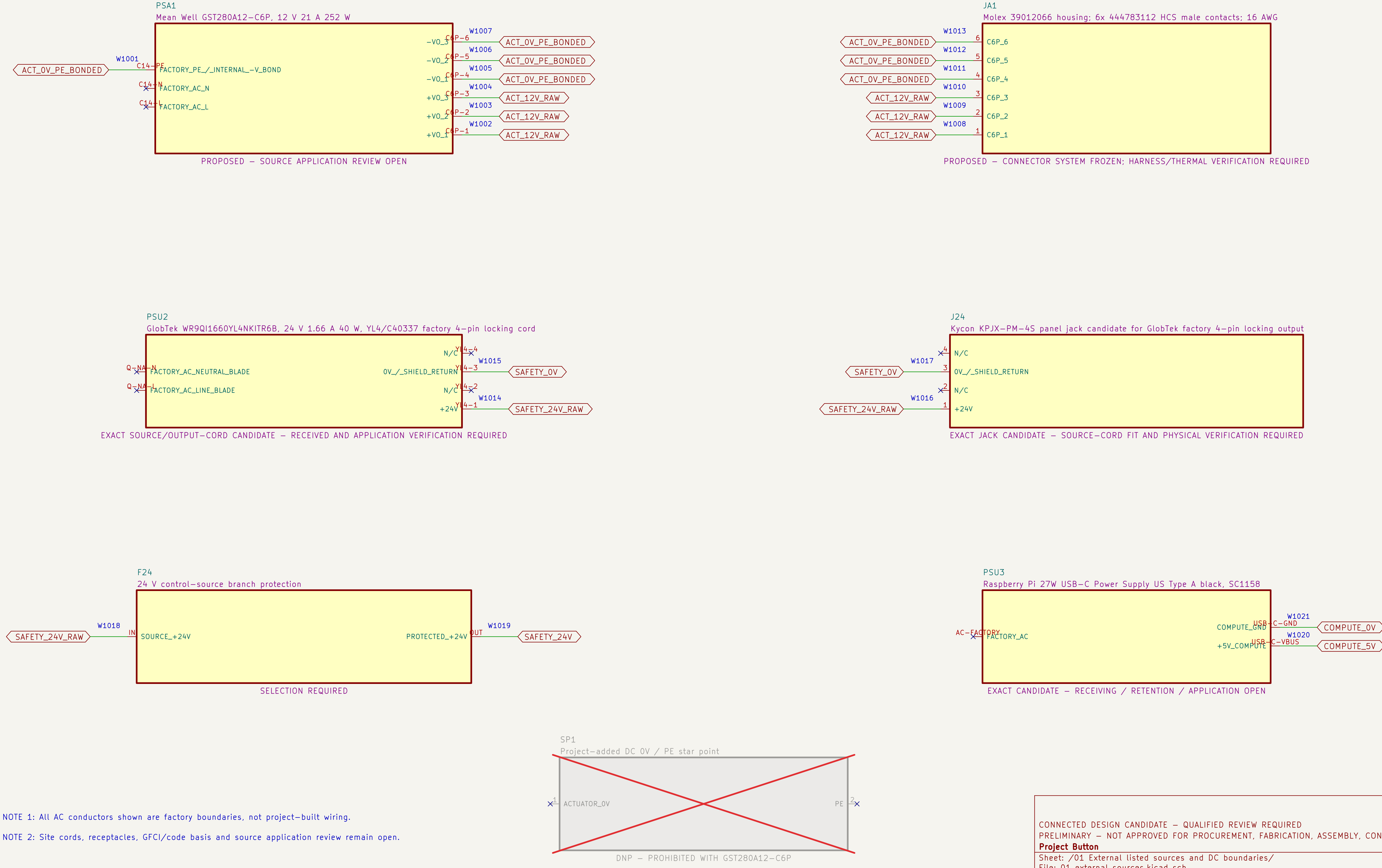
13 Runtime diagnostic observation interfaces

File: 13_runtime_observation_system.kicad_sch

V2.1 PRESERVED; V3 IS A CONNECTED CANDIDATE
PRELIMINARY — NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TE
Project Button
Sheet: /
File: project-button-v3-p1.16-observation-candidate.kicad_sch
Title: PROJECT BUTTON HR-V0 ELECTRICAL V3 OBSERVATION-INTEGRATED CAN
Size: A3 | Date: 2026-08-10 | **Rev: V3-P1.16-OBSERVATION**
KiCad E.D.A. 10.0.5 | Id: 1/14

01 External listed sources and DC boundaries

Factory–sealed adapters eliminate project–built mains wiring.



NOTE 1: All AC conductors shown are factory boundaries, not project–built wiring.

NOTE 2: Site cords, receptacles, GFCI/code basis and source application review remain open.

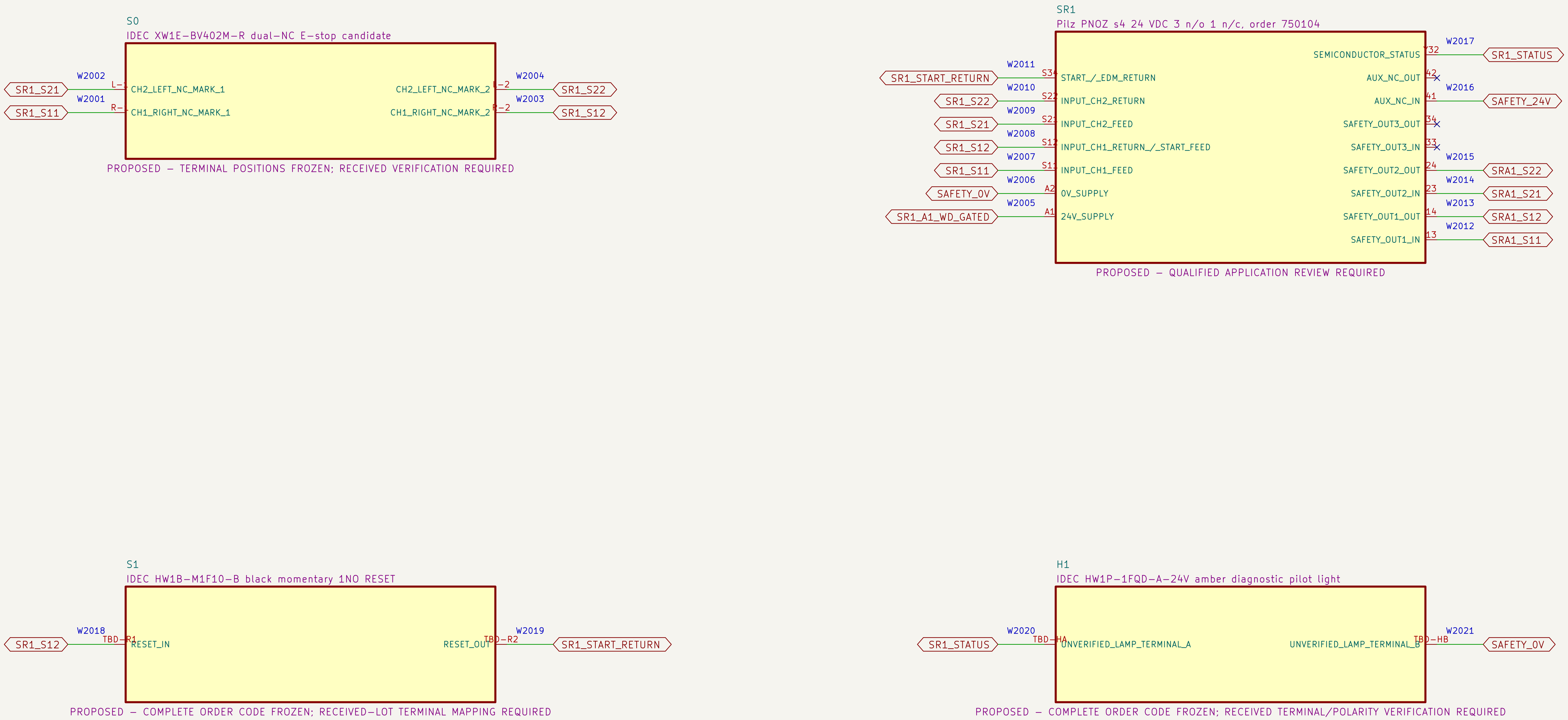
CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TE
Project Button
Sheet: /01 External listed sources and DC boundaries/
File: 01_external_sources.kicad_sch

Title: PB HR–V0 V3–P1.16–OBSERVATION–CANDIDATE – 01

Size: A3 | Date: 2026–08–10 | **Rev: V3–P1.16–OBSERVATION**
KiCad E.D.A. 10.0.5 | Id: 2/14

02 Dual-channel E-stop and RESET eligibility

Each SR1 input return contains only one S0 NC contact; RESET cannot energize K1/K2.



NOTE 1: S0 directly completes both SR1 input returns; ordinary KWD terminals are absent from both loops.

NOTE 2: Heartbeat loss removes SR1 A1 through the separate two-contact supply gate; recovery alone cannot restore the monitored RESET stage.

CONNECTED DESIGN CANDIDATE — QUALIFIED REVIEW REQUIRED
PRELIMINARY — NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TE

Project Button

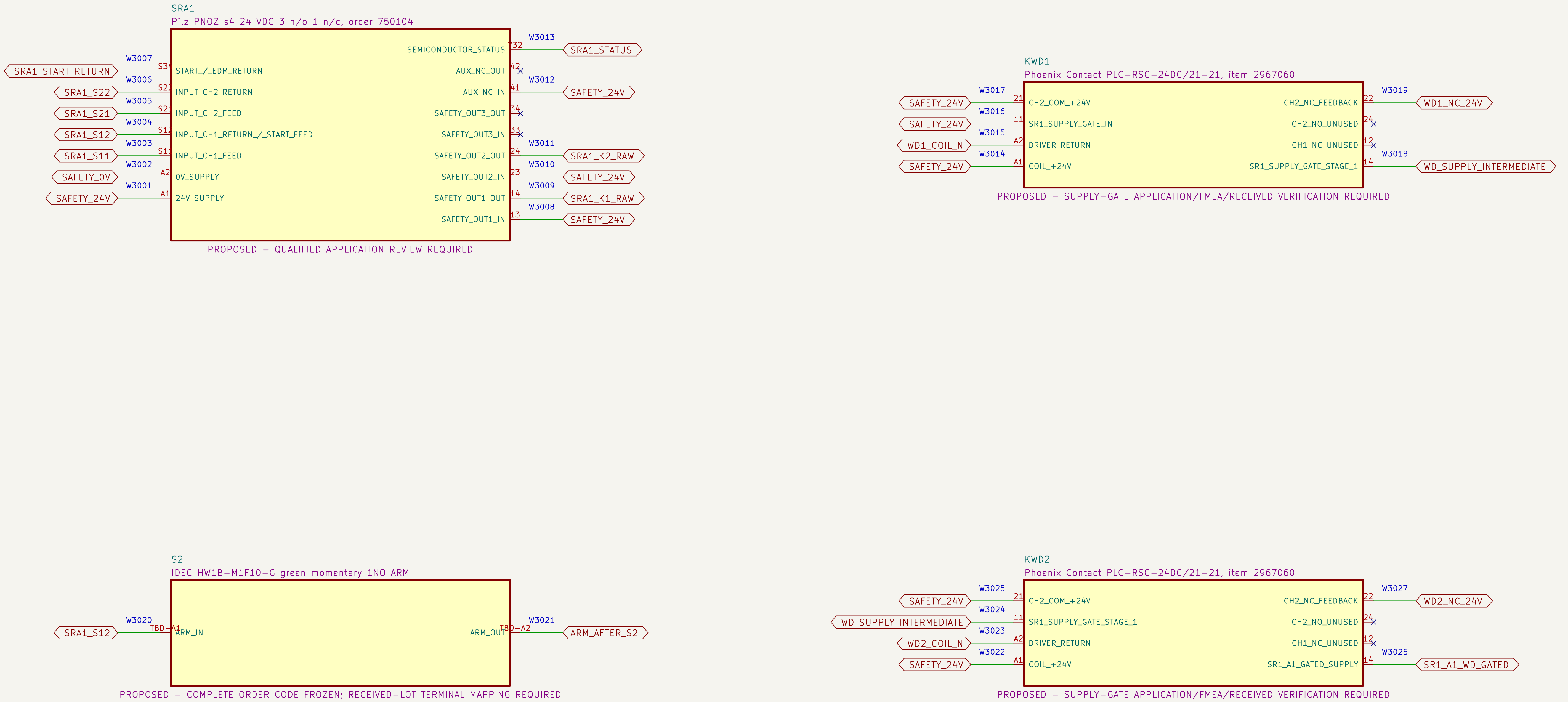
Sheet: /02 Dual-channel E-stop and RESET eligibility/
File: 02_estop_eligibility.kicad_sch

Title: PB HR-V0 V3-P1.16-OBSERVATION-CANDIDATE — 02

Size: A3	Date: 2026-08-10	**Rev: V3-P1.16-OBSERVATION**
KiCad E.D.A. 10.0.5		Id: 3/14

03 Distinct ARM and watchdog–gated SR1 supply

Two ordinary KWD contacts gate SR1 A1; SRA1 still requires SR1 outputs, EDM proof and a new ARM action.



NOTE 1: Required after E–stop/watchdog dropout: cause healthy –> KWD supply restored –>
RESET press/release –> SAFE_READY –> distinct ARM press/release.

NOTE 2: KWD contacts are in series with SR1 A1 only; S0 remains direct in both SR1 input returns and SRA1 receives both SR1 safety outputs directly.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TE

Project Button

Sheet: /03 Distinct ARM and watchdog–gated SR1 supply/
File: 03_arm_watchdog_eligibility.kicad_sch

Title: PB HR–V0 V3–P1.16–OBSERVATION–CANDIDATE – 03

Size: A3 | Date: 2026–08–10 | **Rev: V3–P1.16–OBSERVATION**

KiCad E.D.A. 10.0.5 | Id: 4/14

04 Contactor coils, mirror contacts and EDM

K1 and K2 are distinct final elements; their mirror contacts form the monitored restart return.

NOTE 1: SRA1 outputs are separately protected before K1 and K2 coils. FSR1/FSR2 use proposed Phoenix PT 4–HESI item 3211861 holders with D–ST 4 item 3030420 end cover; both fuse links, grouping and coordination remain SELECTION REQUIRED.

NOTE 2: Loaded DC interruption, suppression behavior, mirror–contact use and coordination require qualified review.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TE

Project Button

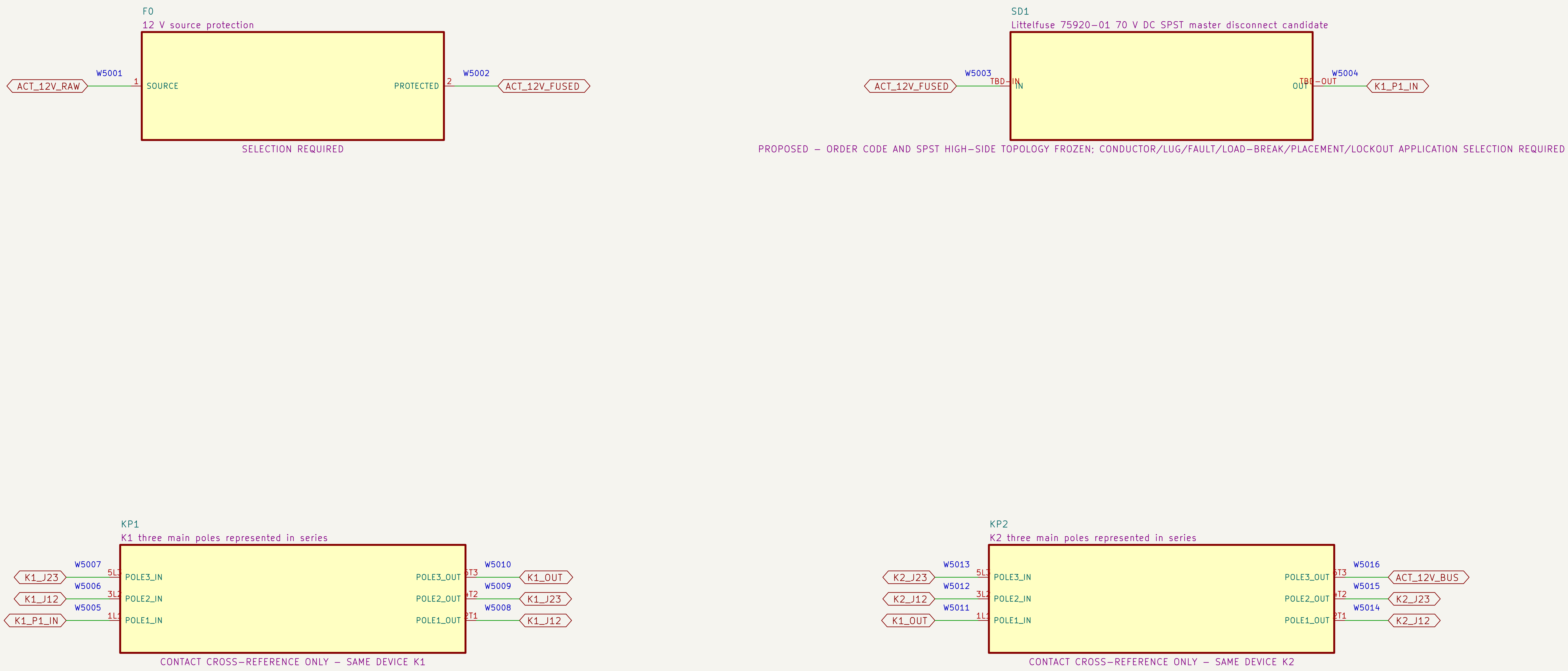
Sheet: /04 Contactor coils, mirror contacts and EDM/
File: 04_contactor_edm.kicad_sch

Title: PB HR–V0 V3–P1.16–OBSERVATION–CANDIDATE – 04

Size: A3	Date: 2026–08–10	**Rev: V3–P1.16–OBSERVATION**
KiCad E.D.A. 10.0.5		Id: 5/14

05 Redundant actuator–power interruption

Source protection, service disconnect and all three poles of K1 then K2 are represented in series.



NOTE 1: Series jumpers: K1 2T1–>3L2, 4T2–>5L3, 6T3–>K2 1L1; repeat through K2 to ACT_12V_BUS.

NOTE 2: No fuse, disconnect, conductor, connector or contactor application is released without fault–current evidence.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED

PRELIMINARY – NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TE

Project Button

Sheet: /05 Redundant actuator–power interruption/

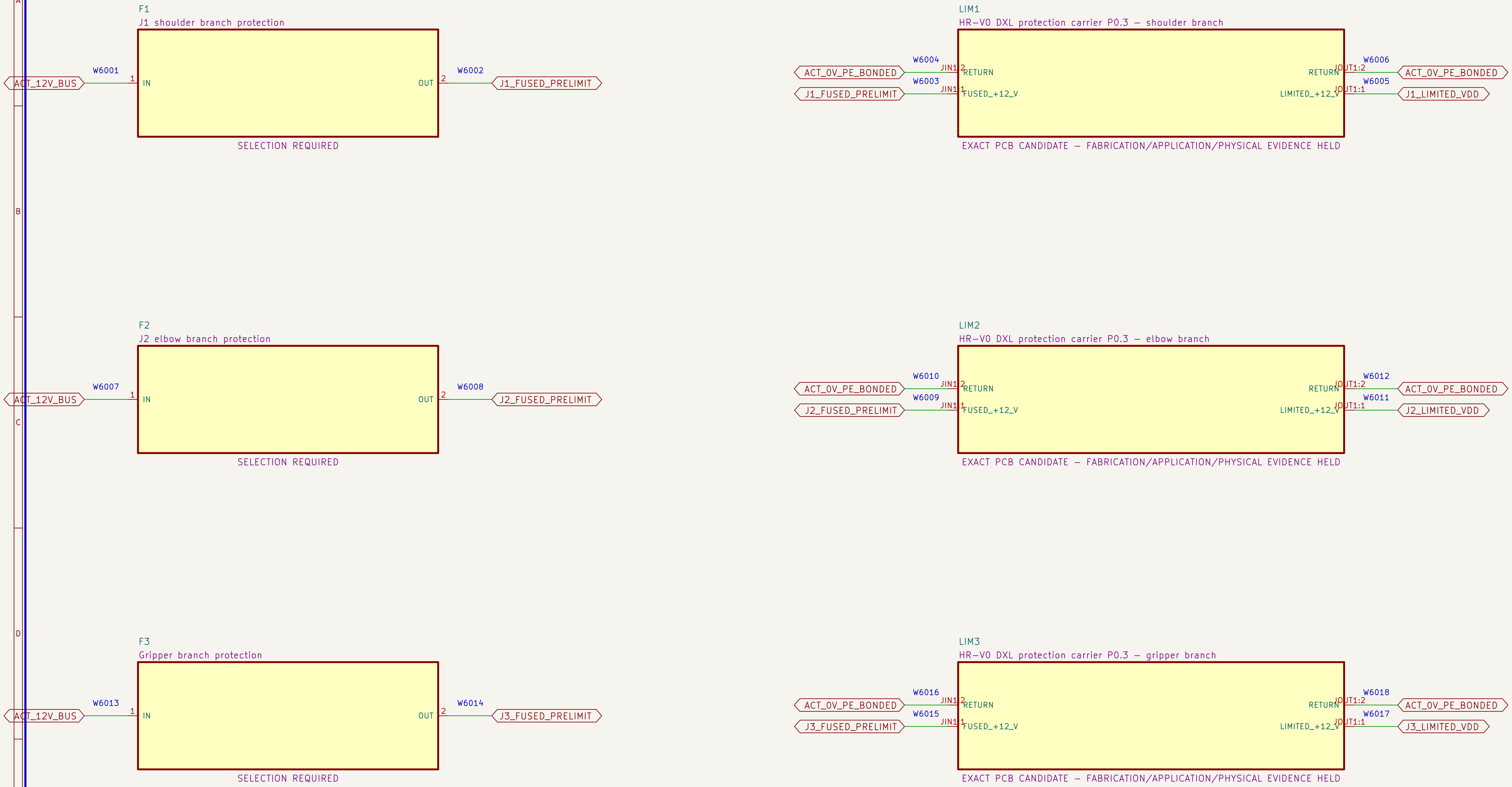
File: 05_actuator_interruption.kicad_sch

Title: PB HR–V0 V3–P1.16–OBSERVATION–CANDIDATE – 05

Size: A3	Date: 2026–08–10	**Rev: V3–P1.16–OBSERVATION**
KiCad E.D.A. 10.0.5		Id: 6/14

06 Protected actuator branches and current–limiter carriers

Each fuse output and carrier output has a distinct positive rail; physical protection performance remains unproved.



NOTE 1: INJ1 is a fixed central star: U2D2 pin 2 is unrouted and J1/J2/J3 positive rails never join.

NOTE 2: Exact actuator cables, branch returns, enclosure, continuity, isolation, signal–integrity, pull and no–backfeed tests remain release gates.

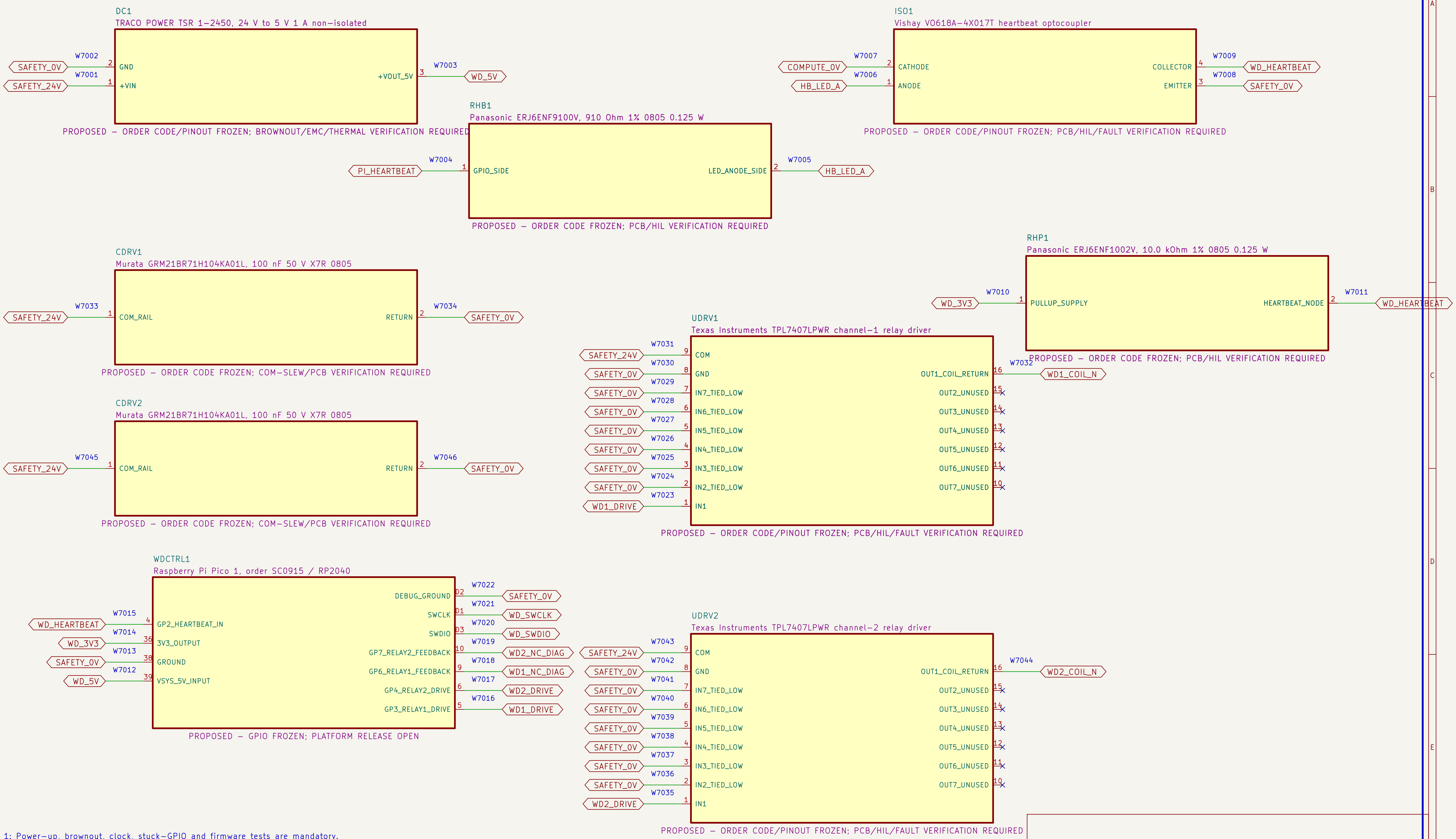
CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TE
Project Button
Sheet: /06 Protected actuator branches and current–limiter carriers/
File: 06_branches_and_limiters.kicad_sch

Title: PB HR–V0 V3–P1.16–OBSERVATION–CANDIDATE – 06

Size: A3 | Date: 2026–08–10 | **Rev: V3–P1.16–OBSERVATION**
KiCad E.D.A. 10.0.5 | Id: 7/14

07 Independent watchdog power, controller and drivers

The ordinary watchdog controller and drivers receive no safety–integrity credit by assertion.



NOTE 1: Power–up, brownout, clock, stuck–GPIO and firmware tests are mandatory.

NOTE 2: Qualified review decides whether watchdog loss is credited or diagnostic only;

current safety credit is NONE.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TE

Project Button

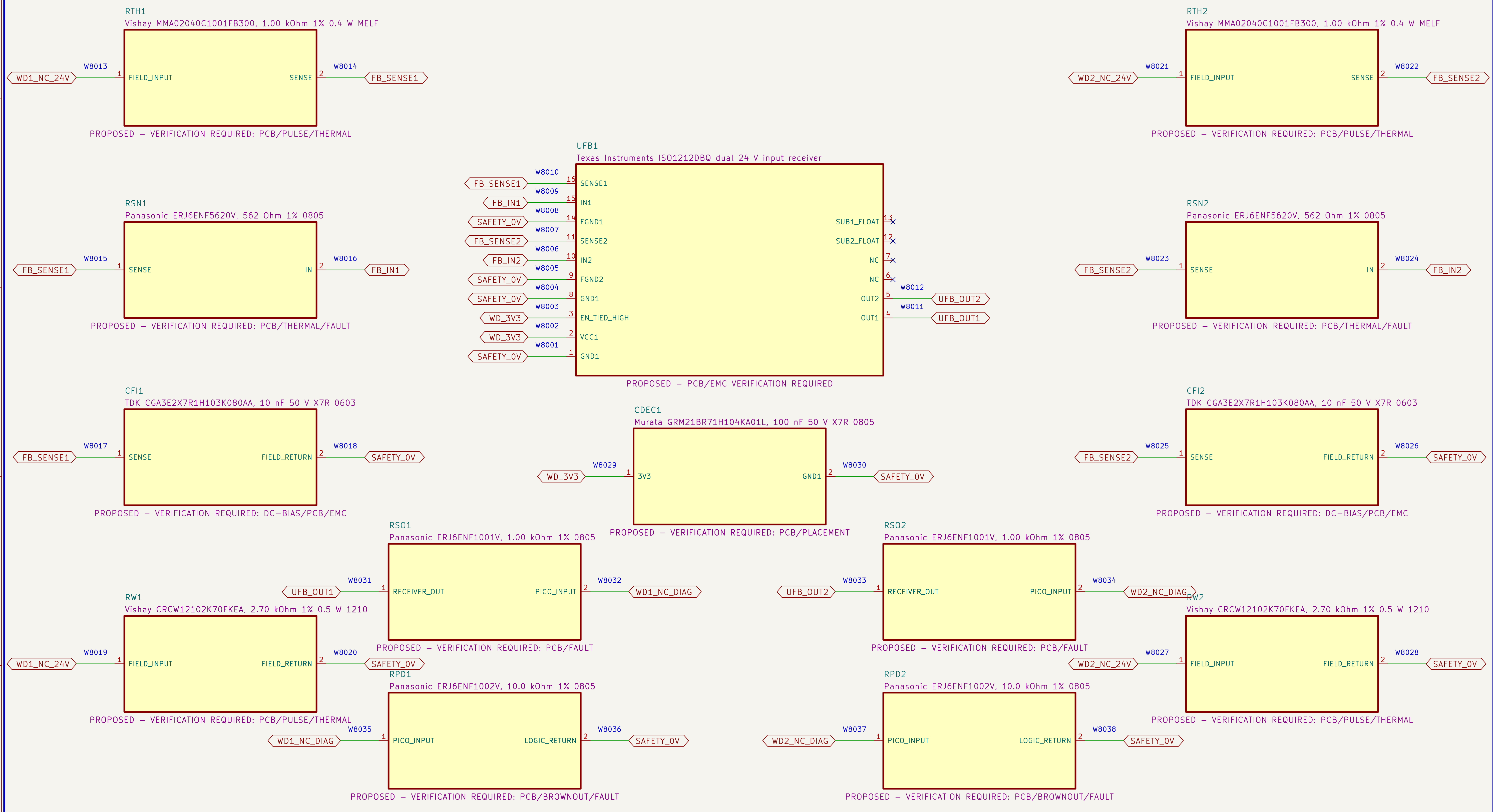
Sheet: /07 Independent watchdog power, controller and drivers/
File: 07_watchdog_control.kicad_sch

Title: PB HR–V0 V3–P1.16–OBSERVATION–CANDIDATE – 07

Size: A3	Date: 2026–08–10	**Rev: V3–P1.16–OBSERVATION**
KiCad E.D.A. 10.0.5		Id: 8/14

08 Calculated dual-channel 24 V watchdog feedback

ISO1212DBQ converts both relay NC diagnostics to default-low 3.3 V logic; no isolation or safety credit is claimed.



NOTE 1: Type-3 values: RTHR=1.00 kOhm, RSENSE=562 Ohm, CIN=10 nF; calculated wetting load is
2.70 kOhm 1% 0.5 W per channel.

NOTE 2: Both grounds are SAFETY_0V; the ISO1212 barrier is not credited. Passive identities
are frozen; PCB, derating, EMC and fault tests remain open.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TE

Project Button

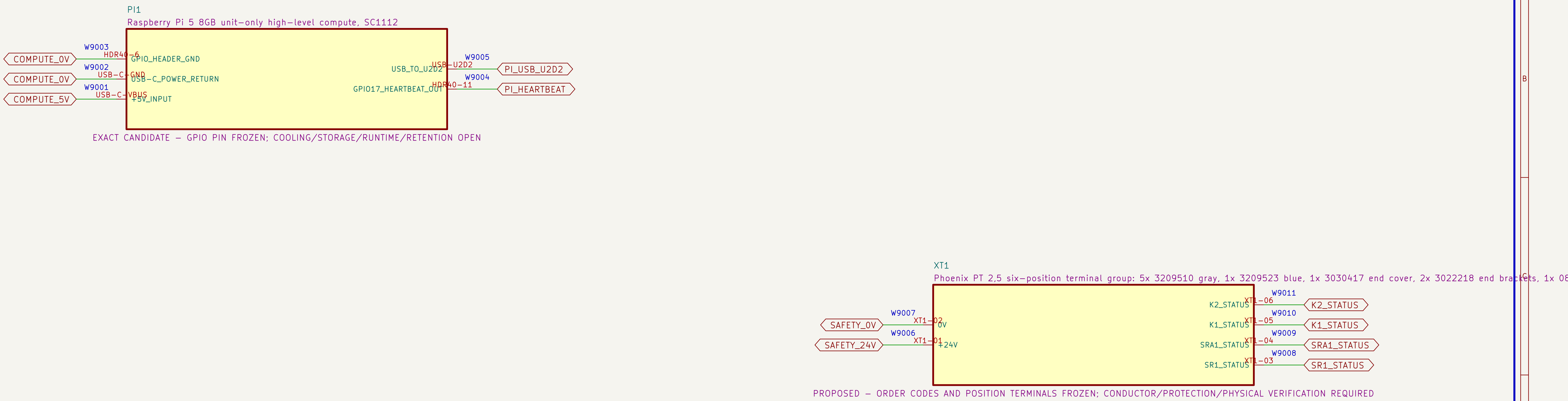
Sheet: /08 Calculated dual-channel 24 V watchdog feedback/
File: 08_watchdog_feedback_interface.kicad_sch

Title: PB HR-V0 V3-P1.16-OBSERVATION-CANDIDATE – 08

Size: A3 | Date: 2026-08-10 | **Rev: V3-P1.16-OBSERVATION**
KiCad E.D.A. 10.0.5 | Id: 9/14

09 Compute and control terminals

High−level compute and diagnostic wiring have no authority to bypass or restore the safety chain.



NOTE 1: No installed debug connector exists; watchdog programming uses only TP15/TP16/TP2 under a future controlled unpowered−fixture procedure.

NOTE 2: Debug connection, halt, flash and disconnect must not enable outputs, back−power the board or bypass a protective function.

NOTE 3: Heartbeat cable, GPIO runtime, timing, terminal family, markers, conductor range, torque and enclosure layout remain selection required.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TE

Project Button

Sheet: /09 Compute and control terminals/
File: 09_compute_and_control_terminals.kicad_sch

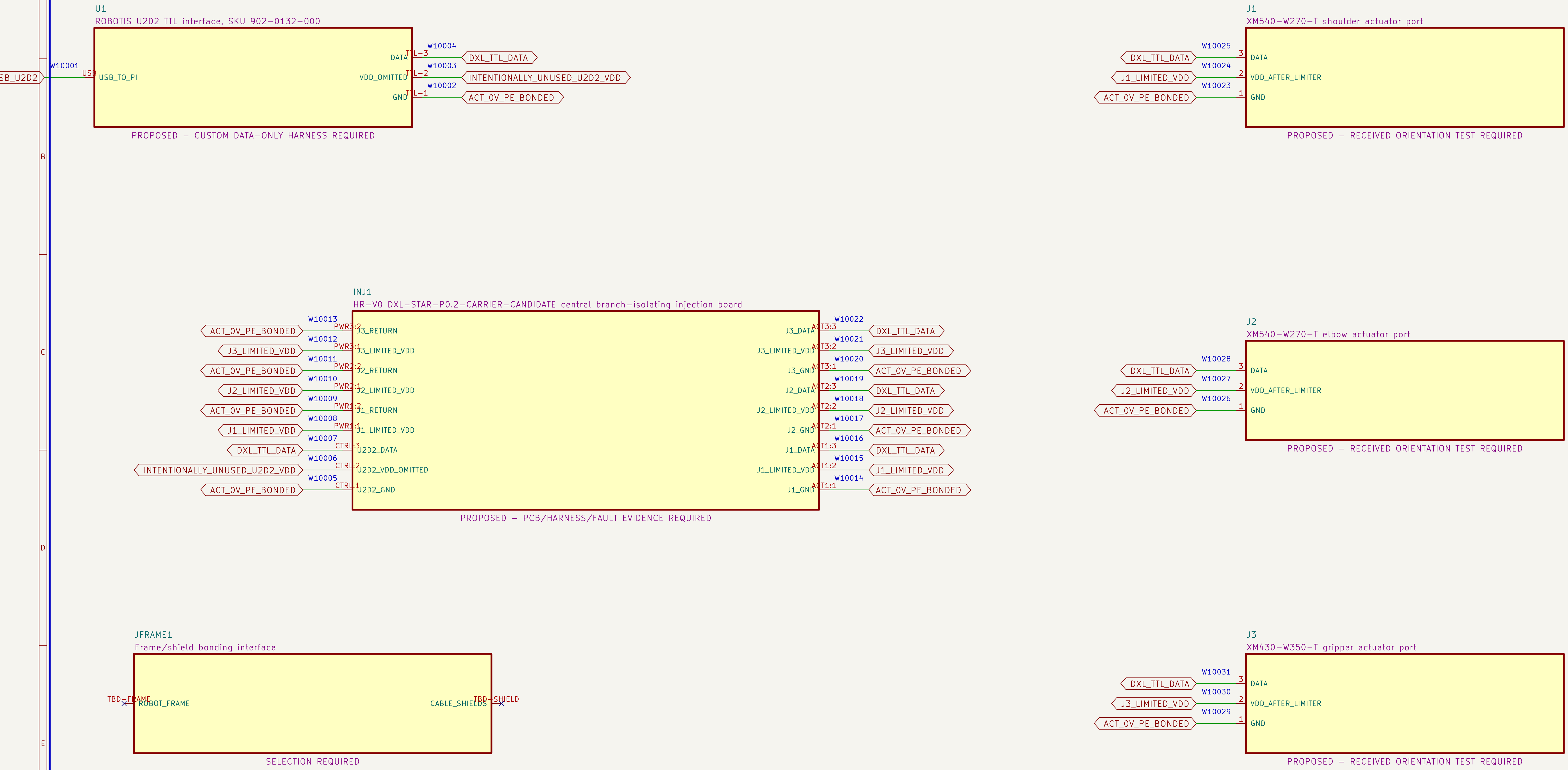
Title: PB HR−V0 V3−P1.16−OBSERVATION−CANDIDATE – 09

Size: A3	Date: 2026−08−10	**Rev: V3−P1.16−OBSERVATION**
KiCad E.D.A. 10.0.5		Id: 10/14

PRELIMINARY – NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TESTING, MOTION, OR ENERGIZATION

10 U2D2, DXL star, actuator ports and bonding boundary

U2D2 pin 2 is omitted; DXL–STAR accepts three distinct limited positive rails and routes DATA/return.



NOTE 1: The V0 bus is TTL; HR–30 RS–485 remains a separate architecture.

NOTE 2: Do not bond frame or shields to 0V/PE until EMC and parallel–path review accepts the exact implementation.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TE

Project Button

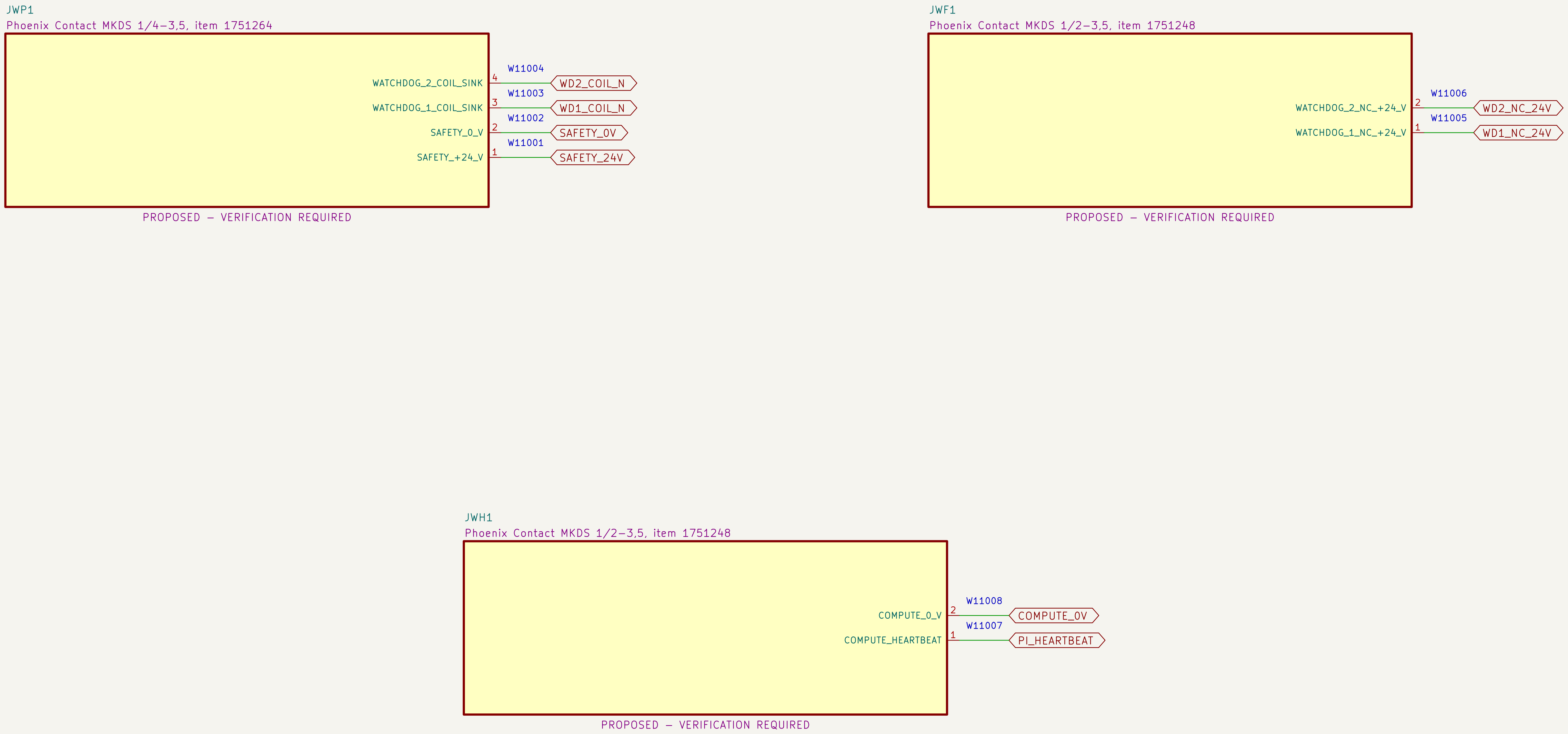
Sheet: /10 U2D2, DXL star, actuator ports and bonding boundary/
File: 10_actuator_interfaces.kicad_sch

Title: PB HR–V0 V3–P1.16–OBSERVATION–CANDIDATE – 10

Size: A3	Date: 2026–08–10	Rev: V3–P1.16–OBSERVATION
KiCad E.D.A. 10.0.5		Id: 11/14

11 Watchdog PCB external connectors

Exact PCB terminal-block candidates and project pin allocations define the board boundary; harness release remains open.



NOTE 1: Terminal numbering follows the PCB footprint and must be confirmed against received parts before wiring.

NOTE 2: Nominal terminal ratings do not release branch protection, conductor, ferrule, harness, enclosure or thermal application.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TE

Project Button

Sheet: /11 Watchdog PCB external connectors/
File: 11_watchdog_pcb_connectors.kicad_sch

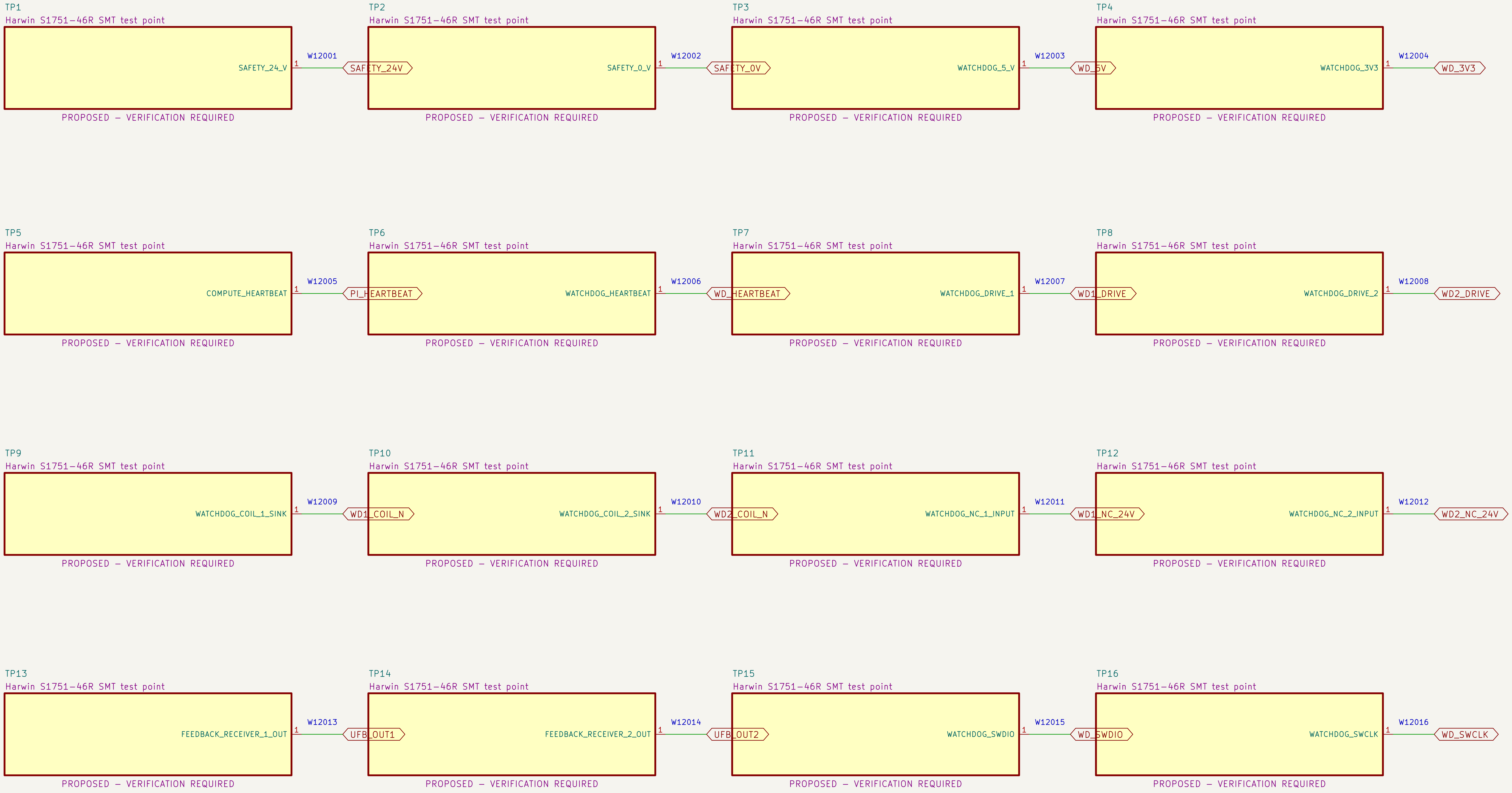
Title: PB HR-V0 V3-P1.16-OBSERVATION-CANDIDATE – 11

Size: A3 | Date: 2026-08-10 | **Rev: V3-P1.16-OBSERVATION**

KiCad E.D.A. 10.0.5 | Id: 12/14

12 Watchdog PCB test access

Exact clip-compatible test terminals expose controlled nodes without granting safety or operating authority.



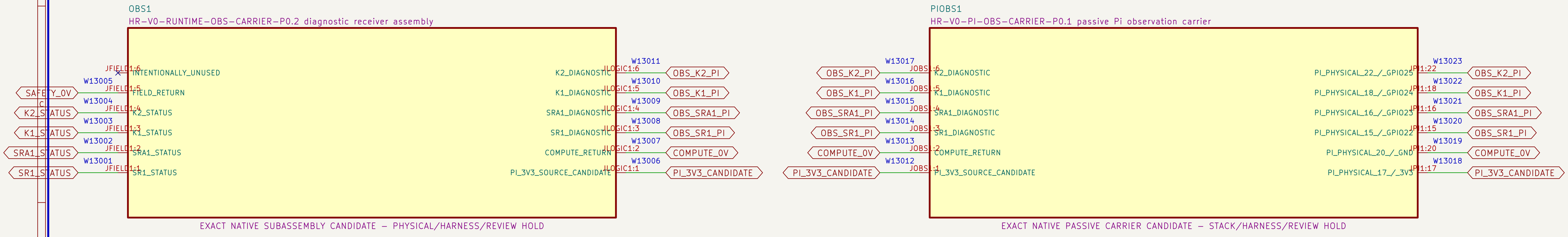
NOTE 1: Test terminals are diagnostic features only; they provide no safety integrity or bypass authority.

NOTE 2: Verify clip clearance with guards, harnesses and power removed before any live measurement procedure is released.

CONNECTED DESIGN CANDIDATE — QUALIFIED REVIEW REQUIRED
PRELIMINARY — NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TE
Project Button
Sheet: /12 Watchdog PCB test access/
File: 12_watchdog_pcb_test_access.kicad_sch
Title: PB HR-V0 V3-P1.16-OBSERVATION-CANDIDATE — 12
Size: A3 | Date: 2026-08-10 | **Rev: V3-P1.16-OBSERVATION**
KiCad E.D.A. 10.0.5 | Id: 13/14

13 Runtime diagnostic observation interfaces

XT1 status conductors feed the R202 isolated receiver; R204 carries six compute−side nets to exact Pi physical pins.



NOTE 1: OBS1 JFIELD1.1−.5 bind directly to XT1−03, −04, −05, −06 and −02 by named net;
JFIELD1.6 is deliberately unused.

NOTE 2: The compute harness binds JLOGIC1.1−.6 to JOBS1.1−.6 one−for−one; Pi physical pins
are 17, 20, 15, 16, 18 and 22.

NOTE 3: Heartbeat GPIO17 on Pi physical pin 11 and return pin 6 remain a separate ordinary
interface on JWH1.

NOTE 4: The observation chain is diagnostic only: it cannot command, restore, latch or

CONNECTED DESIGN CANDIDATE − QUALIFIED REVIEW REQUIRED
PRELIMINARY − NOT APPROVED FOR PROCUREMENT, FABRICATION, ASSEMBLY, CONNECTION, POWERED TE
Project Button
Sheet: /13 Runtime diagnostic observation interfaces/
File: 13_runtime_observation_system.kicad_sch
Title: PB HR−V0 V3−P1.16−OBSERVATION−CANDIDATE − 13
Size: A3 | Date: 2026−08−10 | **Rev: V3−P1.16−OBSERVATION**
KiCad E.D.A. 10.0.5 | Id: 14/14